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PRACTICAL FILE

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These practicals helped me practise working with documents, spreadsheets and databases. Formatting a page, writing a formula and setting up a table now feel much easier than they did at the start of the session.

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01

TASK 1 · LIBREOFFICE BASE

Product Table, Query and Report

TASK

Consider the following Product table:

Tasks:

- Sort the Product table by Price in descending order.
- Use a wildcard to create a query that retrieves all products whose ProductName starts with the letter T.
- Generate a report based on the Product table.
- Format the report to show products in ascending order of ProductID.
- Add the page number in the footer of the report.

Product table

ProductID	ProductName	Category	Price
1	Laptop	Electronics	80000
2	Tablet	Electronics	30000
3	Chair	Furniture	15000
4	Desk	Furniture	25000
5	Smartphone	Electronics	50000

STEPS

Step 1. Open LibreOffice Base and open the database that contains the Product table.

Step 2. Open the Product table and confirm that ProductID, ProductName, Category and Price are present.

Step 3. Sort the Price field in descending order so the highest-priced product appears first.

Step 4. Choose Create Query in Design View. Add the Product table and double-click ProductName.

Step 5. In the Criteria row under ProductName, enter LIKE 'T%' (or LIKE 'T*' if the database uses the * wildcard). Run the query.

Step 6. Save the query with a clear name, such as ProductsStartingWithT.

Step 7. Use the Report Wizard and select the Product table as the source. Include all required fields.

Step 8. Set the report sort field to ProductID in ascending order. Complete the wizard.

Step 9. Edit the report footer and insert a page-number field, then save and preview the report.

RESULT

After sorting Price in descending order, the products are Laptop (₹80,000), Smartphone (₹50,000), Tablet (₹30,000), Desk (₹25,000) and Chair (₹15,000). The wildcard query returns Tablet. Use % for SQL wildcards, or * if that is the wildcard setting in the database. The report sorts ProductID in ascending order and shows page numbers in the footer.

SCREENSHOTS

INSERT SCREENSHOT HERE

Product table in Data View after sorting Price descending; show column headings and all five products.

Fig. 1.1 *Product table in Data View after sorting Price descending*

INSERT SCREENSHOT HERE

Query design/criteria showing the wildcard under ProductName and the query result containing Tablet.

Fig. 1.2 *Query design/criteria showing the wildcard under ProductName and the query result containing Tablet*

INSERT SCREENSHOT HERE

Finished report sorted ProductID 1–5, with a visible page number in the footer.

Fig. 1.3 *Finished report sorted ProductID 1–5, with a visible page number in the footer*

02

TASK 2 · LIBREOFFICE CALC

Compare Two Versions of a Financial Report

TASK

You have received two versions of a financial report from different teams. You need to identify discrepancies and reconcile the differences between the two sheets. How would you do this task in LibreOffice Calc?

The question gives no amounts, so the following sample data is used.

Team A report

Item	Amount
Rent	12000
Groceries	6500
Utilities	2200
Miscellaneous	1800

Team B report

Item	Amount
Rent	12000
Groceries	6800
Utilities	2100
Miscellaneous	1800

Reconciliation sheet

Item	Team A Amount	Team B Amount	Difference	Status
Rent	12000	12000	0	Matched
Groceries	6500	6800	-300	Check
Utilities	2200	2100	100	Check
Miscellaneous	1800	1800	0	Matched

STEPS

Step 1. Open LibreOffice Calc and create three sheets named Team A, Team B and Reconciliation.

Step 2. Make sure both team sheets use the same row order and the same item names. Add headings such as Item, Team A Amount and Team B Amount.

Step 3. Copy or enter the two reports in Team A and Team B. Check that the periods, units and categories match.

Step 4. In Reconciliation, list each item once. Link the Team A amount and Team B amount using cell references.

Step 5. In a Difference column, enter a formula such as =B2-C2, adjusting the cell references to match the layout. Fill the formula down.

Step 6. Add a Status column. Use a formula such as =IF(D2=0;"Matched";"Check") (some locales use commas instead of semicolons).

Step 7. Select the Difference column and apply conditional formatting to highlight non-zero values.

Step 8. Compare flagged items against the source reports and ask the relevant team to verify any mismatch. Record approved corrections and reasons in a Notes column.

Step 9. Save the workbook with all original values preserved and a clear reconciliation summary.

RESULT

A difference of 0 means the amounts match. A positive difference means Team A has the higher amount; a negative difference means Team B has the higher amount. Here Rent and Miscellaneous match, while Groceries (−300) and Utilities (+100) are flagged Check and are verified against the source reports. The original reports are left unchanged and any approved correction is recorded in the Reconciliation sheet.

SCREENSHOTS

INSERT SCREENSHOT HERE

Team A sheet showing its headings and four sample items.

Fig. 2.1 *Team A sheet showing its headings and four sample items*

INSERT SCREENSHOT HERE

Team B sheet showing the matching item names and its amounts.

Fig. 2.2 *Team B sheet showing the matching item names and its amounts*

INSERT SCREENSHOT HERE

Reconciliation sheet with Difference and Status formulas; apply conditional formatting to highlight non-zero differences.

Fig. 2.3 *Reconciliation sheet with Difference and Status formulas*

03

TASK 3 · LIBREOFFICE BASE

Create a Table Using Design View

TASK

Create the following table using Design View:

- Sort the records by Place in descending order.
- Set Roll No. as the primary key.
- Delete record A002.
- Add another record.

Student records

Roll No.	Name	Date of Birth	Phone No.	Place
A001	Manan Gupta	12/09/1987	2354667667	Delhi
A002	Ryansh Sharma	09/05/1986	3435555555	Chattisgarh
A003	Rahul John	02/05/1990	6565544324	Mumbai

STEPS

Step 1. Open LibreOffice Base and choose Tables → Create Table in Design View.

Step 2. Create fields Roll No. (Text), Name (Text), Date of Birth (Date), Phone No. (Text) and Place (Text). Text is suitable for phone numbers because they are identifiers and may contain leading zeroes.

Step 3. Right-click the row selector for Roll No. and choose Primary Key.

Step 4. Save the table as StudentRecords. Open it in Data View and enter the three records exactly as shown in the question.

Step 5. Click the Place column header and choose descending sort. Confirm that the records are ordered by Place from Z to A.

Step 6. Locate record A002 and delete that record only. Confirm the deletion if prompted.

Step 7. Move to the new-record row and enter A004, Kavya Mehta, 15/08/1989, 9876543210, Jaipur.

Step 8. Save the table and reopen it to verify the primary key, sort order and final records.

RESULT

Roll No. is the primary key, so every record has a unique value. After deleting A002, records A001 and A003 remain along with the new record A004 (Kavya Mehta, Jaipur).

SCREENSHOTS

INSERT SCREENSHOT HERE

Design View showing the fields and the key symbol beside Roll No.

Fig. 3.1 *Design View showing the fields and the key symbol beside Roll No*

INSERT SCREENSHOT HERE

Data View after sorting Place descending, before deleting A002 if you want to show the sort operation.

Fig. 3.2 *Data View after sorting Place descending, before deleting A002 if you want to show the sort operation*

INSERT SCREENSHOT HERE

Final table after deleting A002 and adding the approved new record.

Fig. 3.3 *Final table after deleting A002 and adding the approved new record*

04

TASK 4 · LIBREOFFICE BASE

School Database Relationships and Query

TASK

You are managing a school database that includes two main tables: Students and Classes.

- Set up a relationship between the Students table and the Classes table.
- Write a query to list all students along with the class they are in and the teacher of that class.

Students Table

StudentID	StudentName	DateOfBirth	ClassID
1	Priya Sharma	2005-03-15	101
2	Rahul Kapoor	2004-07-22	102
3	Meena Patel	2005-11-10	101
4	Aisha Khan	2004-02-25	103
5	Dev Rai	2005-05-30	102

Classes Table

ClassID	ClassName	TeacherName
101	Mathematics	Mr. Prakash
102	Science	Ms. Yashika
103	History	Mrs. Seema

STEPS

Step 1. Create a database and make a Students table with StudentID as the primary key. Add StudentName, DateOfBirth and ClassID fields with suitable data types.

Step 2. Create a Classes table with ClassID as the primary key, plus ClassName and TeacherName. Enter the supplied data exactly.

Step 3. Open Tools → Relationships. Add Students and Classes to the relationship window.

Step 4. Drag Classes.ClassID onto Students.ClassID to create a one-to-many relationship. Enable referential integrity if the data types and existing values allow it.

Step 5. Create a query in Design View and add both tables.

Step 6. Add StudentName from Students, ClassName from Classes and TeacherName from Classes to the query grid.

Step 7. Check that Students.ClassID is joined to Classes.ClassID. Run the query and verify all five students appear.

Step 8. Save the query as StudentClassTeacherQuery.

RESULT

The query lists each student with the matching class and teacher: Priya Sharma — Mathematics — Mr. Prakash; Rahul Kapoor — Science — Ms. Yashika; Meena Patel — Mathematics — Mr. Prakash; Aisha Khan — History — Mrs. Seema; Dev Rai — Science — Ms. Yashika.

SCREENSHOTS

INSERT SCREENSHOT HERE

Students and Classes tables with their entered records (capture both if they fit clearly).

Fig. 4.1 *Students and Classes tables with their entered records (capture both if they fit clearly)*

INSERT SCREENSHOT HERE

Tools → Relationships window showing Classes.ClassID linked to Students.ClassID.

Fig. 4.2 *Tools → Relationships window showing Classes.ClassID linked to Students.ClassID*

INSERT SCREENSHOT HERE

Query result showing StudentName, ClassName and TeacherName for all five students.

Fig. 4.3 *Query result showing StudentName, ClassName and TeacherName for all five students*

05

TASK 5 · LIBREOFFICE CALC

Spreadsheet Average and Optimization

TASK

Consider the following spreadsheet:

- Calculate the average units sold in cell B8.
- Maximize the average number of units sold to 95.
- Use the following criteria for maximizing the average number of units sold:

Spreadsheet

Row	A	B	C
1	Product	Units Sold	
2	A	69	
3	B	73	
4	C	88	
5	D	86	
6			
7			
8	Average Unit Sold		
9			

Optimization constraints

Cell	Constraint
B2	≤ 88
B3	≥ 82
B4	≤ 90
B5	≥ 60

STEPS

Step 1. Enter the product names in A2:A5 and units sold in B2:B5. Keep the supplied starting values 69, 73, 88 and 86.

Step 2. Click B8 and enter =AVERAGE(B2:B5). Press Enter. The starting average is 79.

Step 3. Open Tools → Solver. Set Target Cell to $B$8 and select Maximize.

Step 4. Set the variable cells to $B$2:$B$5.

Step 5. Add constraints: B2 <= 88, B3 >= 82, B4 <= 90 and B5 >= 60.

Step 6. Add one more constraint, B8 <= 95, so that Solver maximises the average up to the target of 95.

Step 7. Click Solve and inspect the result. B8 becomes 95 while B2 <= 88, B3 >= 82, B4 <= 90 and B5 >= 60 still hold.

Step 8. Keep the solution, check that B8 still contains =AVERAGE(B2:B5), and save the sheet.

RESULT

The starting average is 79. B3 and B5 have no upper limit, so the target of 95 is entered as the constraint B8 <= 95. Solver then finds values whose average is 95; one valid solution is B2 = 88, B3 = 100, B4 = 90 and B5 = 102 (total 380).

SCREENSHOTS

INSERT SCREENSHOT HERE

Sheet with the four starting values and the B8 formula visible in the formula bar.

Fig. 5.1 *Sheet with the four starting values and the B8 formula visible in the formula bar*

INSERT SCREENSHOT HERE

Solver dialog showing target $B$8, Maximize, variable cells $B$2:$B$5 and the four supplied constraints.

Fig. 5.2 *Solver dialog showing target $B$8, Maximize, variable cells $B$2:$B$5 and the four supplied constraints*

INSERT SCREENSHOT HERE

Final Solver result with the average in B8 equal to 95.

Fig. 5.3 *Final Solver result with the average at 95*

06

TASK 6 · LIBREOFFICE WRITER

Create a Bank Cash-Withdrawal Slip Template

TASK

Create a pro forma for a bank cash-withdrawal slip and add it to the Templates section.

STEPS

Step 1. Open Writer and create a new document. Type a centred heading such as BANK CASH WITHDRAWAL SLIP.

Step 2. Insert a table with fields for date, account holder name, account number, branch, amount in figures, amount in words, signature of account holder and authorised signature. Leave blank spaces for later use.

Step 3. Adjust table column widths, borders and spacing so the form fits on one page. Add a small bank name/logo placeholder only if permitted.

Step 4. Choose File → Templates → Save as Template. Name it Cash Withdrawal Slip and save it under My Templates.

RESULT

The bank cash-withdrawal slip is saved as a reusable template under My Templates.

SCREENSHOTS

INSERT SCREENSHOT HERE

Finished one-page withdrawal slip with the heading and all fields visible.

Fig. 6.1 *Finished one-page withdrawal slip with the heading and all fields visible*

INSERT SCREENSHOT HERE

Save as Template dialog showing the template name Cash Withdrawal Slip and My Templates category.

Fig. 6.2 *Save as Template dialog showing the template name Cash Withdrawal Slip and My Templates category*

INSERT SCREENSHOT HERE

Template visible in the template manager or a new document opened from the saved template.

Fig. 6.3 *Template visible in the template manager or a new document opened from the saved template*

07

TASK 7 · LIBREOFFICE WRITER

Create a New Year Greetings Template

TASK

Create a New Year greetings template with the salutation on the left and a picture in the right-hand corner.

STEPS

Step 1. Create a second Writer document for New Year greetings. Use a two-column table or text boxes to keep the salutation on the left and the picture at the upper-right.

Step 2. Type a greeting such as “Happy New Year!” and add a short message. Insert a suitable picture using Insert → Image.

Step 3. Set the image anchor and text wrapping so it stays in the right-hand corner without covering the text.

Step 4. Save this document as a template named New Year Greetings. Reopen the New Year Greetings template from File → New → Templates to test it.

RESULT

The New Year greetings document is saved as a template. Reopen it from File → New → Templates to check that the salutation remains on the left and the picture stays in the right-hand corner.

SCREENSHOTS

INSERT SCREENSHOT HERE

Greeting layout with salutation on the left and image in the upper-right.

Fig. 7.1 *Greeting layout with salutation on the left and image in the upper-right*

INSERT SCREENSHOT HERE

Image selected with the placement/wrapping controls visible, if practical.

Fig. 7.2 *Image selected with the placement/wrapping controls visible, if practical*

INSERT SCREENSHOT HERE

Save as Template dialog or the saved New Year Greetings template reopened from Templates.

Fig. 7.3 *Save as Template dialog or the saved New Year Greetings template reopened from Templates*

08

TASK 8 · LIBREOFFICE WRITER

Conservation of Water: Paragraph and Picture

TASK

Write a paragraph on “Conservation of Water” and add a related picture. Resize the image to width 3.77 and height 3.99 (units are not specified in the source question). Crop the picture using scaling parameters of 50% width and 50% height.

STEPS

Step 1. Open Writer and type the heading “Conservation of Water”. Apply a clear heading style.

Step 2. Write a paragraph explaining why water is important and how people can reduce waste, repair leaks, reuse water where safe, harvest rainwater and protect groundwater.

Step 3. Choose Insert → Image and select a relevant water-conservation picture.

Step 4. Right-click the picture and open Properties → Type. Untick Keep ratio and enter width 3.77 and height 3.99 (inches).

Step 5. Open the Crop settings. Apply 50% scaling for width and 50% scaling for height as requested. Preview the crop to ensure the main subject remains visible.

Step 6. Set the image alignment and text wrapping so the paragraph and picture fit neatly on the page.

Step 7. Save the file and check the print preview for clipping or overflow.

RESULT

Paragraph used: “Water is important for people, animals and plants. We can save water by turning off taps, repairing leaks, collecting rainwater and using only the water we need. We should also keep rivers and lakes clean and protect groundwater for future generations.” The picture is sized 3.77 × 3.99 inches and cropped with 50% width and 50% height scaling.

SCREENSHOTS

INSERT SCREENSHOT HERE

Heading and paragraph entered in Writer.

Fig. 8.1 *Heading and paragraph entered in Writer*

INSERT SCREENSHOT HERE

Image Properties/Crop settings showing width 3.77, height 3.99 and the 50% crop/scaling values.

Fig. 8.2 *Image Properties/Crop settings showing width 3.77, height 3.99 and the 50% crop/scaling values*

INSERT SCREENSHOT HERE

Finished page with the paragraph and resized image clearly visible.

Fig. 8.3 *Finished page with the paragraph and resized image clearly visible*

09

TASK 9 · LIBREOFFICE WRITER

Create a Paragraph Style Named “My list”

TASK

Create a new paragraph style with Bookman Old Style font, blue font colour, red text highlight, 18-point size and bold text. Save the style as My list.

Style settings

Setting	Value
Style name	My list
Font	Bookman Old Style
Font style	Bold
Font size	18 pt
Font colour	Blue
Text highlight	Red

STEPS

Step 1. Open Writer and press F11 to open the Styles sidebar.

Step 2. Select Paragraph Styles. Right-click in the list and choose New.

Step 3. On the General tab, enter the style name My list.

Step 4. On the Font tab, choose Bookman Old Style, Bold, and 18 pt.

Step 5. On the Font Effects tab, choose blue as the font colour. Set the text highlight/background to red in the relevant highlighting or character-background control available in your LibreOffice version.

Step 6. Click OK to save the style.

Step 7. Type the sample text “Keep your work neat and organised.”, select the paragraph and double-click My list in the Styles sidebar.

Step 8. Check that the font, size, bold formatting, blue font colour and red highlight are applied. Save the document.

RESULT

The new paragraph style is saved as My list with Bookman Old Style, bold 18-point text, blue font and red highlighting. In some LibreOffice versions, text highlighting is set separately from font colour.

SCREENSHOTS

INSERT SCREENSHOT HERE

New Paragraph Style dialog showing the name My list.

Fig. 9.1 *New Paragraph Style dialog showing the name My list*

INSERT SCREENSHOT HERE

Style properties showing Bookman Old Style, Bold, 18 pt, blue font and red highlight/background settings.

Fig. 9.2 *Style properties showing Bookman Old Style, Bold, 18 pt, blue font and red highlight/background settings*

INSERT SCREENSHOT HERE

Sample sentence with the saved style visibly applied.

Fig. 9.3 *Sample sentence with the saved style visibly applied*

10

TASK 10 · LIBREOFFICE CALC

Subtotal Marks by Subject

TASK

Mr. Laksh is preparing a list for a class of 30 students. He enters marks for five different subjects. Group the students by subject and calculate the subtotals for each subject.

The question gives no marks, so the following sample data (30 students, five subjects) is used.

Student Name	Subject	Marks	Student Name	Subject	Marks
Student 01	English	78	Student 16	Mathematics	95
Student 02	English	82	Student 17	Mathematics	68
Student 03	English	75	Student 18	Mathematics	88
Student 04	English	88	Student 19	Science	81
Student 05	English	69	Student 20	Science	87
Student 06	English	91	Student 21	Science	74
Student 07	Hindi	80	Student 22	Science	89
Student 08	Hindi	76	Student 23	Science	77
Student 09	Hindi	84	Student 24	Science	93
Student 10	Hindi	72	Student 25	Social Science	76
Student 11	Hindi	90	Student 26	Social Science	83
Student 12	Hindi	86	Student 27	Social Science	90
Student 13	Mathematics	92	Student 28	Social Science	71
Student 14	Mathematics	85	Student 29	Social Science	85
Student 15	Mathematics	79	Student 30	Social Science	80

STEPS

Step 1. For student marks, create columns Student Name, Subject and Marks. Enter all 30 student records and ensure every subject is spelled consistently.

Step 2. Sort the data by Subject in ascending order using Data → Sort. This is important because Calc Subtotals works on grouped, sorted records.

Step 3. Choose Data → Subtotals. Under Group by, select Subject; under Calculate subtotals for, select Marks; choose Sum. Click OK.

Step 4. Check that a subtotal row appears under every subject and that a Grand Total appears at the end. Use the outline buttons on the left to collapse or expand the student rows.

RESULT

Subtotals: English 483, Hindi 488, Mathematics 507, Science 501, Social Science 485. Grand total: 2464.

SCREENSHOTS

INSERT SCREENSHOT HERE

Raw 30-row table with Student Name, Subject and Marks.

Fig. 10.1 *Raw 30-row table with Student Name, Subject and Marks*

INSERT SCREENSHOT HERE

Data → Subtotals dialog showing Group by Subject and Sum for Marks.

Fig. 10.2 *Data → Subtotals dialog showing Group by Subject and Sum for Marks*

INSERT SCREENSHOT HERE

Finished grouped sheet with each subject's subtotal rows visible.

Fig. 10.3 *Finished grouped sheet with each subject's subtotal rows visible*

11

TASK 11 · LIBREOFFICE CALC

Analyze Household Expense Scenarios

TASK

Analyze different scenarios of monthly household expenses. Consider Minimum Expense, Average Expense and Maximum Expense for categories such as Rent, Groceries, Utilities and Miscellaneous.

The question gives no amounts, so the following sample values are used.

Category	Minimum Expense	Average Expense	Maximum Expense
Rent	11000	12000	13000
Groceries	4000	5000	6500
Utilities	1500	2200	3000
Miscellaneous	1000	2000	3500
TOTAL	**17500**	**21200**	**26000**

STEPS

Step 1. For household expenses, create columns Category, Minimum Expense, Average Expense and Maximum Expense. Enter the scenario values supplied by your teacher or source worksheet.

Step 2. Use =SUM(B2:B5), =SUM(C2:C5) and =SUM(D2:D5) to calculate each scenario total, adjusting columns to match the layout. Compare the totals to see the lowest, typical and highest monthly expense.

Step 3. Format the amounts as currency and, if required, insert a simple column chart comparing the three totals.

RESULT

The totals are ₹17,500 (minimum), ₹21,200 (average) and ₹26,000 (maximum), so the monthly expense can range from ₹17,500 to ₹26,000.

SCREENSHOTS

INSERT SCREENSHOT HERE

Expense table with four categories and the three scenario columns.

Fig. 11.1 *Expense table with four categories and the three scenario columns*

INSERT SCREENSHOT HERE

Formula bar showing a total formula such as =SUM(B2:B5), with the total row visible.

Fig. 11.2 *Formula bar showing a total formula such as =SUM(B2:B5), with the total row visible*

INSERT SCREENSHOT HERE

Final total row with all three scenario totals; optional comparison chart if you created one.

Fig. 11.3 *Final total row with all three scenario totals*

12

TASK 12 · LIBREOFFICE CALC

Calculate Monthly Product Profit

TASK

A store sells three products (Product X, Product Y and Product Z) at different selling prices and costs. Calculate the monthly profit using multiple operations and considering different sales volumes.

The question gives no prices, costs or volumes, so the following sample data is used.

Product	Selling Price	Cost per Unit	Units Sold	Revenue	Total Cost	Profit
Product X	500	320	80	40000	25600	14400
Product Y	750	500	60	45000	30000	15000
Product Z	300	180	120	36000	21600	14400
TOTAL				**121000**	**77200**	**43800**

STEPS

Step 1. For product profit, create columns Product, Selling Price, Cost per Unit, Units Sold, Revenue, Total Cost and Profit.

Step 2. For each product, calculate Revenue with =SellingPrice*UnitsSold, Total Cost with =CostPerUnit*UnitsSold and Profit with =Revenue-TotalCost. Use cell references in formulas, not typed-in repeated totals.

Step 3. Calculate total monthly profit by summing the product profit cells. Try different sales-volume scenarios by changing the units sold and recording the resulting profit.

Step 4. Record the selling prices, unit costs and units sold for each product using the figures in the worksheet. Compare different sales-volume scenarios without changing the formulas.

RESULT

With the sample data, total revenue is ₹121,000, total cost is ₹77,200 and total profit is ₹43,800. When the units sold change to 90, 70 and 130, the same formulas give a total profit of ₹49,300.

SCREENSHOTS

INSERT SCREENSHOT HERE

Product table with prices, unit costs and units sold entered.

Fig. 12.1 *Product table with prices, unit costs and units sold entered*

INSERT SCREENSHOT HERE

*Formula bar showing a formula such as =B2*D2 and the calculated revenue/cost/profit values.*

Fig. 12.2 *Formula bar showing a formula such as =B2*D2 and the calculated revenue/cost/profit values*

INSERT SCREENSHOT HERE

Total row showing profit for all products; capture the second sales-volume scenario if required.

Fig. 12.3 *Total row showing profit for all products*

13

TASK 13 · LIBREOFFICE CALC

Goal Seek: Monthly Gadget Revenue

TASK

Vishal manages a store that sells gadgets A, B and C at ₹100, ₹150 and ₹200 per unit. He expects to sell 200 units of A and 150 units of B. Determine how many units of C are needed to reach total revenue of ₹50,000 using Goal Seek.

Gadget prices

Gadget	Selling price per unit
A	₹100
B	₹150
C	₹200

STEPS

Step 1. Create columns Gadget, Selling Price, Units and Revenue. Enter A: ₹100 and 200 units; B: ₹150 and 150 units; C: ₹200 and leave its units as a starting guess.

Step 2. Calculate revenue for each row using =B2*C2, adjusted to your sheet layout.

Step 3. Calculate total revenue with =SUM(D2:D4).

Step 4. Choose Tools → Goal Seek. Set Formula Cell to the total revenue cell. Set Target Value to 50000. Set Variable Cell to the units cell for Gadget C.

Step 5. Click OK and accept the result if Goal Seek finds a solution. Format the revenue cells as currency.

Step 6. Verify the result manually and save the workbook.

RESULT

Gadget A brings in ₹20,000 and Gadget B brings in ₹22,500, for a total of ₹42,500. The remaining amount is ₹7,500. At ₹200 per unit, Gadget C needs to sell 37.5 units. If only whole units are allowed, 37 units gives ₹49,900 and 38 units gives ₹50,100.

SCREENSHOTS

INSERT SCREENSHOT HERE

Gadget table with formula cells and total revenue cell D5 visible.

Fig. 13.1 *Gadget table with formula cells and total revenue cell D5 visible*

INSERT SCREENSHOT HERE

Goal Seek dialog showing Formula Cell D5, Target Value 50000 and Variable Cell C4.

Fig. 13.2 *Goal Seek dialog showing Formula Cell D5, Target Value 50000 and Variable Cell C4*

INSERT SCREENSHOT HERE

Final sheet showing 37.5 for Gadget C and total revenue ₹50,000.

Fig. 13.3 *Final sheet showing 37.5 for Gadget C and total revenue ₹50,000*

14

TASK 14 · LIBREOFFICE CALC

Add Hyperlinks to a Book Inventory

TASK

The school librarian wants to include a hyperlink in her book inventory spreadsheet that directs students to the school library's online catalogue. Help her add the hyperlink.

Book inventory

Book Title	Online Catalog Link
To Kill a Mockingbird	[Link to Catalog]
1984	[Link to Catalog]
The Great Gatsby	[Link to Catalog]

STEPS

Step 1. Open Calc and create the two columns Book Title and Online Catalog Link. Enter the three titles exactly as supplied.

Step 2. Ask the librarian for the official school library catalogue URL. The question does not provide the actual URL, so do not invent one.

Step 3. Select the link cell for the first title and press Ctrl+K, or choose Insert → Hyperlink.

Step 4. Choose Internet or Web, paste the official catalogue URL and set the display text to Open library catalogue. Click Apply/OK.

Step 5. Repeat for each row. If the same catalogue page is used for all titles, the same URL can be used; otherwise use the correct title-specific catalogue links supplied by the librarian.

Step 6. Test each link by Ctrl-clicking or using the hyperlink action, depending on the Calc settings. Save the spreadsheet.

RESULT

The sheet shows all three titles, each with a clickable link labelled "Open library catalogue" that opens the school library's online catalogue.

SCREENSHOTS

INSERT SCREENSHOT HERE

Book inventory table with all three titles and the link display-text column.

Fig. 14.1 *Book inventory table with all three titles and the link display-text column*

INSERT SCREENSHOT HERE

Hyperlink dialog showing the official catalogue URL and display text.

Fig. 14.2 *Hyperlink dialog showing the official catalogue URL and display text*

INSERT SCREENSHOT HERE

Finished sheet with clickable link text visible for all three books.

Fig. 14.3 *Finished sheet with clickable link text visible for all three books*

15

TASK 15 · LIBREOFFICE CALC

Consolidate Data from Two Sheets

TASK

Consolidate the data from the two sheets below into a third sheet in LibreOffice Calc.

Sheet 1

Name	June Pocket Money
Amit	1200
Sumit	1000
Naina	1500
Suman	1300
Kamal	1400

Sheet 2

Name	July Pocket Money
Amit	900
Sumit	1100
Naina	1300
Suman	1200
Kamal	1250

STEPS

Step 1. Create three sheets named Sheet1, Sheet2 and Consolidated.

Step 2. On Sheet1, enter the Name and June Pocket Money columns exactly as shown. On Sheet2, enter Name and July Pocket Money.

Step 3. On Consolidated, create columns Name, June Pocket Money, July Pocket Money and Total. Enter each name once in the same row order.

Step 4. Link the June values from Sheet1 and July values from Sheet2 using cell references, for example =Sheet1.B2 and =Sheet2.B2. Use the exact sheet names and cell locations in your workbook.

Step 5. In the Total column, enter =SUM(B2:C2) and fill the formula down for all five names.

Step 6. Check that every name matches between the two source sheets and that no row has shifted. Format amounts as currency if required.

Step 7. Save the workbook.

RESULT

The totals are Amit 2,100, Sumit 2,100, Naina 2,800, Suman 2,500 and Kamal 2,650. The June total is 6,400 and the July total is 5,750, giving a combined total of 12,150.

SCREENSHOTS

INSERT SCREENSHOT HERE

Sheet1 with the June data.

Fig. 15.1 *Sheet1 with the June data*

INSERT SCREENSHOT HERE

Sheet2 with the July data.

Fig. 15.2 *Sheet2 with the July data*

INSERT SCREENSHOT HERE

Consolidated sheet showing linked values, row totals and the grand total.

Fig. 15.3 *Consolidated sheet showing linked values, row totals and the grand total*

16

TASK 16 · LIBREOFFICE BASE

Create and Modify the Salesman Table

TASK

Create a table named Salesman with the following structure:

Salesman fields

Field Name	Data Type
SM_ID	Number (NUMERIC)
SM_Name	Text (35 Characters)
Address1	Text (40 Characters)
Address2	Text (40 Characters)
City	Text (15 Characters)
State	Text (15 Characters)
PinCode	Number
Sell_Amt	Number
Comm	Number
Phone	Number

The question gives no records, so the following 15 fictional sample records are used. Gender fills with Male by default.

SM_ID	SM_Name	Address1	Address2	City	State	PinCode	Sell_Amt	Comm	Phone
1	Arjun Sharma	12 Green Park	Block A	Delhi	Delhi	110016	45000	5	9876501001
2	Neha Verma	23 Model Town	Block B	Delhi	Delhi	110009	52000	6	9876501002
3	Rohan Gupta	14 Sector 14	Near Market	Gurugram	Haryana	122001	61000	7	9876501003
4	Priya Singh	8 Civil Lines	Lane 2	Jaipur	Rajasthan	302006	47000	5	9876501004
5	Aman Khan	19 Ring Road	Floor 1	Lucknow	Uttar Pradesh	226001	39000	4	9876501005
6	Kavya Patel	31 MG Road	Flat 2	Ahmedabad	Gujarat	380001	56000	6	9876501006
7	Dev Mehta	6 Lake View	Main Road	Mumbai	Maharashtra	400001	72000	8	9876501007
8	Sana Ali	44 Park Street	Block C	Kolkata	West Bengal	700016	43000	5	9876501008
9	Rahul Das	17 Station Road	Near Post Office	Patna	Bihar	800001	41000	4	9876501009
10	Anaya Rao	29 Beach Road	Unit 4	Chennai	Tamil Nadu	600001	65000	7	9876501010
11	Kabir Joshi	12 Lake Road	Block A	Pune	Maharashtra	411001	58500	6	9876501011
12	Meera Nair	7 Hill View	Near School	Kochi	Kerala	682001	51000	6	9876501012
13	Ishaan Roy	15 Market Road	Unit 8	Bhopal	Madhya Pradesh	462001	48000	5	9876501013
14	Diya Sethi	22 Central Avenue	Block D	Chandigarh	Chandigarh	160017	63000	7	9876501014
15	Yash Kapoor	9 Rose Garden	Floor 3	Dehradun	Uttarakhand	248001	46000	5	9876501015

STEPS

Step 1. Open LibreOffice Base and choose Tables → Create Table in Design View.

Step 2. Create the fields exactly as listed: SM_ID, SM_Name, Address1, Address2, City, State, PinCode, Sell_Amt, Comm and Phone. Set the specified data types and text lengths.

Step 3. Set SM_ID as the primary key if required by the database design, then save the table as Salesman.

Step 4. Open the table in Design View again and add a field named Gender. Set its default value to Male. Save the table.

Step 5. Open Salesman in Data View and enter the 15 sample records. Confirm that Gender fills with Male automatically.

Step 6. Delete the records with SM_ID 3 (Rohan Gupta) and SM_ID 6 (Kavya Patel), as instructed.

Step 7. Return to Design View. Change Address1 and Address2 from Text(40) to Text(50). Save the changes.

Step 8. Sort the SM_Name field in ascending order and inspect the result. Then sort it in descending order and inspect the result.

Step 9. Save and reopen the table to confirm the field widths, default Gender value, record count and sort behaviour.

RESULT

> After the two deletions, 13 records remain. Address1 and Address2 are Text (50) and Gender defaults to Male. Sorted A–Z the list starts with Aman Khan; sorted Z–A it starts with Yash Kapoor.

SCREENSHOTS

INSERT SCREENSHOT HERE

Design View with the fields and the Gender default value Male; show Address1/Address2 updated to Text (50).

Fig. 16.1 *Design View with the fields and the Gender default value Male*

INSERT SCREENSHOT HERE

Data View sorted by SM_Name ascending after removing the two records; show Gender values as Male.

Fig. 16.2 *Data View sorted by SM_Name ascending after removing the two records*

INSERT SCREENSHOT HERE

Data View sorted by SM_Name descending after the same deletions.

Fig. 16.3 *Data View sorted by SM_Name descending after the same deletions*

Conclusion

In this file, I practised using LibreOffice Writer, Calc and Base. I created templates and paragraph styles in Writer. In Calc, I used formulas, hyperlinks, subtotals, Goal Seek, Solver and financial reconciliation. In Base, I created tables, set keys and relationships, ran queries and prepared reports.

In Writer, I made reusable templates for a bank withdrawal slip and a New Year greeting, sized and cropped a picture to exact measurements, and saved a paragraph style that can be applied to any heading in one click.

In Calc, I linked two sheets with cell references, used Goal Seek to find that Gadget C needs 37.5 units to reach ₹50,000, and used Solver to push an average up to 95 within the given limits. Sorting marks by subject and adding subtotals showed me how well-organised data makes a summary easy.

In Base, I built tables with primary keys, joined Students and Classes through a relationship, and used a wildcard query to pick out the products that start with T. Changing the Salesman table after entering records taught me to plan field sizes before typing the data.

I also learned to check source data, use clear field names, verify formulas and keep original records when reconciling differences. Where a question gave no data, I used clearly labelled sample data.

These activities helped me become more confident with document formatting, spreadsheets and databases.

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